DhanChetna (Working Title): A Context-Aware Expense Tracking System for Students

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***Abstract*—Financial management remains a persistent challenge among college students due to irregular income sources, dependency-based financial structures, and impulsive micro-spending behavior. Existing personal finance applications are primarily designed for salaried professionals and do not incorporate contextual modeling tailored to student lifestyles. This paper proposes the design and development of a context-aware expense tracking system specifically targeted at college students. The system integrates rule-based behavioral analysis with machine learning techniques to classify expenses, detect recurring spending patterns, and predict budget exhaustion. The architecture follows a modular monolith design using Django, PostgreSQL, and Firebase Authentication to ensure scalability and maintainability. The proposed system aims to bridge the gap between traditional expense tracking and intelligent financial awareness systems for students.**

***Index Terms*—Expense Tracking, Context-Aware Systems, Student Finance, Budget Prediction, Machine Learning, Financial Analytics**

I. Introduction

Financial literacy and spending discipline are critical life skills. However, most college students operate under irregular income conditions such as pocket money, stipends, or part-time earnings. Traditional financial management tools focus heavily on banking integration, investment tracking, and credit-based analytics, which are less relevant to students.

This project proposes a student-centric expense tracking system that incorporates contextual modeling and behavioral analysis to improve financial awareness.

Proposed application names: DhanChetna/PocketSense, Hisaab Karo, DhanChetna

II. Literature Survey

A. Student-Oriented Expense Tracking Applications

Recent studies highlight the growing adoption of expense tracking applications among university students. A study published in JOIREM [1] examines the efficiency and adoption of expense tracker applications in enhancing students' financial management. The research emphasizes that visual dashboards, budgeting alerts, and spending summaries significantly improve financial awareness and discipline among students. However, the study focuses primarily on user adoption and behavioral outcomes rather than system intelligence or contextual modeling.

Similarly, various open-source student expense tracking systems available on GitHub [2], [3] provide basic functionalities such as transaction logging, category-based budgeting, and visualization dashboards. These systems demonstrate practical implementations of student-focused budgeting tools. However, they largely rely on static categorization and lack advanced features such as behavioral expense classification, predictive analytics, and contextual advisory mechanisms.

B. OCR-Based Expense Tracking Systems

Advancements in automation have introduced Optical Character Recognition (OCR) into expense tracking systems. A study published in the International Journal of Artificial Intelligence [4] presents the development of a student expense tracking system integrated with OCR technology to extract transaction details from receipts. The system reduces manual entry effort and enhances usability.

Another related work [5] further explores OCR-assisted tracking mechanisms to improve financial data capture accuracy. While these systems successfully automate receipt processing, they primarily focus on data acquisition rather than behavioral analytics, spending prediction, or personalized advisory systems.

C. Machine Learning in Financial Applications

Machine learning has been widely applied in financial systems for fraud detection, anomaly detection, and expenditure prediction. Techniques such as Logistic Regression, Random Forest, and Isolation Forest have demonstrated effectiveness in structured financial datasets [6]. Time-series forecasting models such as ARIMA are commonly used to predict expenditure trends [7].

Recent IEEE research [8] explores AI-powered financial management systems integrating machine learning models to generate intelligent financial insights. These systems focus on predictive analytics and smart recommendation engines. However, most implementations are designed for general consumers or banking environments rather than student-specific financial contexts.

D. Open-Source Financial Tracking Architectures

The Django-based financial tracking system by Darshan et al. [9] demonstrates full-stack expense tracking using Django, PostgreSQL, and visualization libraries. The repository provides insights into scalable backend architecture and modular design patterns.

Additionally, repositories under the GitHub topic "budget-tracker" [10] showcase diverse implementations incorporating modern web stacks and occasionally machine learning modules. However, these implementations typically emphasize technical stack integration rather than contextual behavioral modeling.

E. Identified Research Gap

Based on the reviewed literature and open-source systems, the following gaps are identified:

- Most systems provide descriptive analytics rather than contextual or prescriptive insights.
- Student lifestyle factors (hostel living, dependency on pocket money, food availability) are rarely modeled.
- Hybrid systems combining rule-based reasoning and machine learning for financial advisory remain underexplored.
- Limited research addresses repetitive micro-spending detection specific to students.

Therefore, there exists a need for a context-aware student-centric expense tracking system that integrates behavioral classification, predictive budgeting, and contextual advisory mechanisms within a scalable architecture.

III. Dataset Description

The dataset for the DhanChetna/PocketSense system will consist of a hybrid financial dataset constructed using publicly available CC0-licensed reference datasets along with statistically guided synthetic data generation. Instead of directly merging public datasets, their statistical properties will be used as reference points to generate realistic and privacy-preserving financial data. This approach will allow controlled experimentation while avoiding the use of sensitive personal financial records.

A. Data Source

Two publicly available datasets will be used as statistical references:

- *Hostel Life Daily Expense Tracker (Jan–Mar 2025)* [11]
- *Indian Personal Finance and Spending Habits* [12]

The Hostel Life Daily Expense Tracker dataset is licensed under CC BY-SA 4.0 [11], while the Indian Personal Finance dataset is released under the CC0 Public Domain license [12]. Both licenses permit academic use with proper attribution.

These datasets will not be directly merged. Instead, their category distributions, demographic proportions, and spending ranges will be analyzed. Based on these insights, a structured synthetic dataset will be generated programmatically to match the schema of the DhanChetna/PocketSense backend.

B. Dataset Structure

The final dataset will contain two primary components:

1. User Profile Data

Each simulated user profile will include:

- Age
- Study year (if applicable)
- Living situation (hostel, home, rented, PG)
- Scholarship status
- Part-time job status
- Income amount and frequency

These attributes will influence transaction generation and spending behavior.

2. Transaction-Level Data

Each transaction record will include:

- User ID
- Title and description
- Amount
- Category
- Transaction date
- Derived features such as day of week

The dataset will simulate multiple months of activity per user in order to capture recurring expenses, periodic income, and spending variability.

C. Behavioral Characteristics

The generated data will model realistic financial patterns such as:

- Frequent low-value food expenses for hostel students,
- Regular transport expenses for commuting users,
- Monthly subscription payments,
- Increased discretionary spending during weekends.

Transaction amounts will follow a log-normal distribution to better reflect real-world financial skewness instead of uniform randomness.

D. Preprocessing and Feature Engineering

Before training machine learning models, the following preprocessing steps will be applied:

- Text normalization for transaction titles and descriptions,
- TF-IDF vectorization for text features,
- Scaling of numerical features such as amount,
- Extraction of date-based features (weekday, monthly position).

The dataset will be divided into training and validation sets while ensuring that user-level separation is maintained to prevent data leakage.

E. Rationale for Dataset Design

Public financial datasets often lack contextual demographic attributes or realistic student spending behavior. Additionally, privacy concerns limit the availability of real banking transaction data.

By generating a statistically anchored synthetic dataset, the system will allow safe experimentation, scalable machine

learning training, and alignment with the backend database schema, while maintaining ethical and legal standards.

F. Expected Dataset Size

The dataset is expected to include:

- 300–500 simulated users,
- Approximately 20,000–50,000 transactions,
- 3–6 months of simulated activity per user.

This size will be sufficient for training and evaluating the expense categorization and pattern detection components of the DhanChetna/PocketSense system.

IV. Methods and Models

A. System Overview

DhanChetna/PocketSense will be designed as a context-aware expense tracking system aimed primarily at students and young individuals managing limited financial resources. The system will consist of a frontend interface (desktop/web) connected to a Django-based backend API, which will manage business logic, database operations, and machine learning components.

The overall workflow of the system will be as follows:

1) The user will register and complete a financial profile (income type, living situation, scholarship status, etc.).
2) The user will record income and expense transactions through the interface.
3) All transactions will be stored securely in a relational database.
4) The backend will compute real-time summaries such as daily, monthly, and yearly totals.
5) Budget monitoring logic will evaluate spending against predefined limits.
6) Machine learning and rule-based modules will analyze transaction patterns and generate contextual financial advice.

The system will follow a modular architecture where business logic remains in the backend, and the frontend will function as a presentation layer.

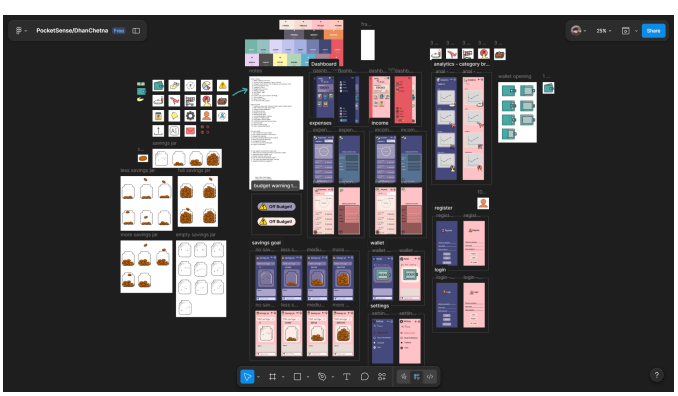

Fig. 1: Application User Interface Design (Version 2)

B. Core Functional Modules

1) Authentication and User Profiling: User authentication will be handled using a secure authentication mechanism. After login, users will complete a profile including:

- Income amount and frequency,
- Living situation (hostel, home, rented, etc.),
- Scholarship or part-time job status,
- Preferred budget limits.

This contextual information will be used to personalize advice and spending analysis.

2) Transaction Management: The system will allow users to:

- Add, edit, and delete expenses,
- Record income entries,
- Assign categories to transactions,
- View transaction history.

Each transaction will include a title, description, amount, category, and date. Real-time validation will ensure correct data entry.

3) Budget Monitoring: Users will be able to set:

- Overall monthly spending limits,
- Category-specific budget limits.

The system will continuously calculate budget utilization and trigger warnings when spending exceeds predefined thresholds. Visual indicators such as progress bars and alerts will be used to enhance user awareness.

4) Savings Goal Module: The application will allow users to define savings goals by specifying:

- Target amount,
- Target date.

Progress toward the goal will be calculated automatically based on remaining balance and user contributions.

5) Analytics and Visualization: The system will generate analytical insights including:

- Weekly and monthly spending summaries,
- Category-wise distribution charts,
- Income versus expense comparisons,
- Trend analysis over time.

These visualizations will help users understand their financial behavior patterns.

C. Machine Learning Models

The system will integrate lightweight machine learning components to enhance financial awareness.

1) Expense Categorization Model: A supervised learning model will be trained using TF-IDF features extracted from transaction titles and descriptions. A Logistic Regression classifier will be used to predict transaction categories when the user does not manually specify one.

2) Recurring Expense Detection: Recurring expenses will initially be detected using rule-based frequency analysis. Transactions with similar titles and amounts occurring at regular intervals will be grouped as recurring patterns.

3) Context-Aware Advice Engine: A hybrid rule-based system will generate personalized advice based on:

- Spending ratios,
- Income level,
- Living situation,
- Repetitive expense detection.

For example, if food expenses exceed a high percentage of monthly income for a hostel student, the system will generate an awareness-based recommendation.

D. Data Flow and Processing

All transaction data will be processed in the backend. Upon each transaction entry:

1) The transaction will be validated and stored in the database.
2) Budget utilization will be recalculated.
3) Recurring pattern detection logic will be triggered.
4) Advice summaries will be updated.

This modular processing design will ensure scalability and maintainability.

E. Design Philosophy

The system will prioritize:

- Simplicity in user interaction,
- Context-aware financial insights,
- Privacy-preserving data handling,
- Lightweight and interpretable machine learning models.

The primary goal will be to provide actionable financial awareness rather than complex financial forecasting.

V. Conclusion

This project proposes DhanChetna/PocketSense, a context-aware expense tracking system designed primarily for students and young individuals managing limited financial resources. The system aims to combine structured financial tracking with lightweight machine learning and rule-based analysis to provide personalized financial insights rather than generic budgeting advice.

The proposed architecture emphasizes modular backend design, interpretable models, and privacy-preserving data handling. A statistically guided synthetic dataset will be developed to support experimentation and model training while avoiding the use of sensitive financial data.

Overall, the project establishes a feasible and scalable foundation for implementing a student-focused financial awareness system, which will be further developed and evaluated in subsequent milestones.

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